Project: MobiGuide (Market Driven Phone Recommender)

Literature Review

Smartphone buyers often face difficulty choosing the best device because prices and specifications vary across e-commerce platforms. Smartphone comparison and recommendation systems help users evaluate devices based on factors such as price, camera, battery, and performance.

MobiGuide applies this concept to the Egyptian smartphone market using data collected from Amazon Egypt, Noon Egypt, and Jumia Egypt. The project cleans and standardizes the collected data and uses Power BI to compare smartphones and platforms.

MobiGuide also provides recommendation scores for four user priorities: Gaming, Camera, Battery, and Value-for-Money. It helps users find suitable smartphones based on their budget and needs while comparing recorded prices across different platforms.

The project uses a one-time collected dataset, so the dashboard represents a snapshot of the market and does not provide live prices or availability.

1. Feedback & Evaluation

The MobiGuide project can be evaluated according to how effectively it transforms raw e-commerce smartphone data into useful information for consumers.

The evaluation should focus on the quality of data collection and cleaning, the accuracy of smartphone specifications, the reliability of cross-platform model matching, the logic of the recommendation scores, and the usability of the Power BI dashboard.

Special attention should be given to the Gaming, Camera, Battery, and Value-for-Money scoring methods. These scores should be clearly documented so that the relationship between smartphone specifications and recommendation results can be understood.

The dashboard should also be evaluated by testing different budgets and user priorities to ensure that filters, visualizations, DAX measures, and recommendation rankings work correctly.

2. Suggested Improvements

Several improvements could enhance MobiGuide in future versions:

Automated Data Updates: Replace the one-time data collection process with scheduled data collection where technically and legally appropriate, allowing prices and availability to be updated more frequently.

More E-Commerce Platforms: Expand the dataset by including additional trusted Egyptian smartphone retailers.

Advanced Recommendation Logic: Improve the scoring model by allowing users to assign their own importance weights to specifications.

Historical Price Analysis: Store data from multiple collection dates to analyze price changes over time.

Improved Model Matching: Develop stronger matching rules for identifying the same smartphone when product names differ between platforms.

User Testing: Collect feedback from potential users to improve dashboard navigation, recommendation presentation, and overall usability.

Additional Smartphone Factors: Include other relevant factors when reliable data is available, such as warranty, charging speed, chipset performance, display technology, and software support.

These improvements could transform MobiGuide from a static analytical project into a more comprehensive smartphone decision-support system.

3. Final Grading Criteria

The final project evaluation can be divided into four major areas:

Documentation

The documentation should clearly explain the project problem, objectives, scope, data sources, scraping methodology, data-cleaning process, data model, recommendation formulas, DAX calculations, risks, limitations, and final findings.

Implementation

Implementation should evaluate the successful completion of the main project components, including data collection, data cleaning and transformation, price-tier

classification, Gaming/Camera/Battery/Value-for-Money scoring, cross-platform price comparison, and the six-page Power BI dashboard.

Testing

Testing should verify data quality, calculations, DAX measures, cross-platform matching, recommendation rankings, dashboard filters, slicers, and interactions. Different budgets and user priorities should be tested to confirm that the dashboard produces consistent and meaningful results.

Presentation

The final presentation should clearly communicate the market problem, project methodology, data-processing workflow, recommendation approach, dashboard functionality, major insights, project limitations, and potential future improvements. The team should also demonstrate the Power BI dashboard and explain how MobiGuide can help users make more informed smartphone purchasing decisions.